

# Causal discovery with parameter analysis

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## Abstract

abstract

## 1 Introduction

## 2 The main algorithm

There given a sample set of interventions

$$V_{\text{do}} = \{\text{do}(V = v_0), \text{do}(V = v_1), \dots\},$$

where  $v_i$  (is a) set of all nodes.

1. Assume we have samples of  $V_{\text{do}}$  (with «a» distribution) and the variables for any  $V \in v$

$$S_1^v, \dots, S_n^v.$$

2. For any  $y \in V$ , compute  $\text{Anc}(y)$  by:

$$\text{Anc}(y) = \{v, y \not\subseteq V_{\text{do}}\}.$$

3. For each,  $\forall Y \in V, \forall x \in \text{Anc}(y)$ :

- (a) estimate  $w_i^{yx} = \arg \max_w P(y \mid X_{\text{do}}; w)$ ,  $1 \leq i \leq n$  and obtain  $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$ , where  $P(y \mid X_{\text{do}}; w)$  in the probability of  $Y$  from  $\text{Anc}(Y)$  based on all samples, assessed with  $X$ ;
- (b) compute  $\hat{\boldsymbol{\mu}}^{yx}$  and  $\hat{\boldsymbol{\Sigma}}^{yx}$  from  $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$ ;
- (c) compute  $D_{\text{KL}}\left(p(w; \hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) \parallel p(w; \hat{\boldsymbol{\mu}}^{xy}, \hat{\boldsymbol{\Sigma}}^{xy})\right)$ ;
- (d) if  $D_{\text{KL}}(\parallel) \geq 0$ ,  $x \in p_a(Y)$ .

### The first variant

$$X_{\text{do}}\{\text{do}(X = x_0), \text{do}(X = x_1), \dots\};$$

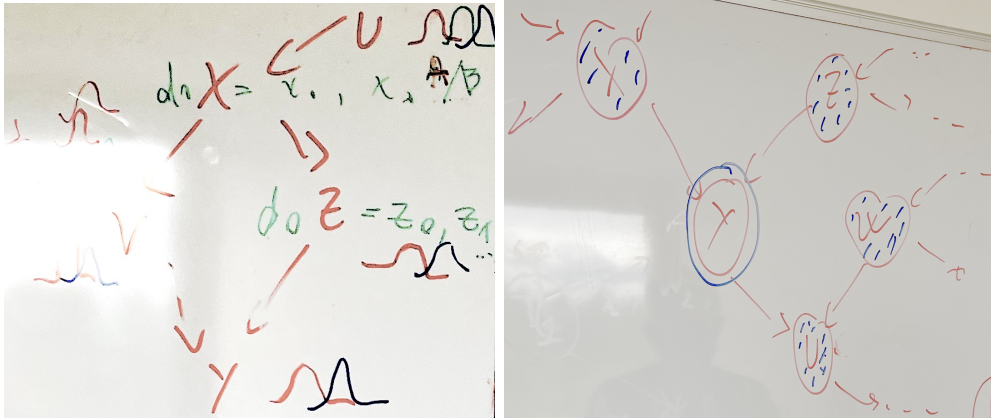
$V_i$  set of all nodes

(a) If  $X_{\text{do}} \not\perp\!\!\!\perp Y$ , then  $X \in \text{Anc}(Y)$

- If there exists  $\text{do}(Z)$ , where  $Z = V \setminus X$ , s.t.  $X \perp\!\!\!\perp Y \mid \text{do}(Z)$ , then  $X \notin \text{Pa}(Y)$ .
- Otherwise,  $X \in \text{Pa}(Y) \rightarrow$  impossible in practice

(b) With (a) we can compute the ancestors of all variables , e.q.

$$\begin{aligned} \text{Anc}(Y) = \{V, Z, X, V\}, \text{Anc}(X) = \{V\} &\Rightarrow \text{Pa}(X) = \{V\} \\ \text{Anc}(V) = \{X, V\}, \text{Anc}(V) = \{\emptyset\} &\Rightarrow \text{Pa}(V) = \{\emptyset\} \\ \text{Anc}(Z) = \{X, V\}, &\Rightarrow ?Z \perp\!\!\!\perp X \mid V \end{aligned}$$



## 3 Classification as a criterion of causal disturbance

- Set a tests for the causal disturbance:  $X$  and  $X = \mathbf{x}_0$ .
- Train a classifier to distinguish between the parameter distribution
- The distribution of the model parameters are the features.
- Maximize the gap between two classes

## Introduction

The goal of this document is to describe the target function of the causal graph.

We treat the algorithm PCGCE in the paper *Extended summary graphs (ESG)* belongs to class of the exhaustive search algorithms. So to an optimal causal graph we make no difference between

- 1) exhaustive printing or combinatorial algorithms,
- 2) gradient-based algorithms including algorithms with relaxations,
- 3) zero-gradient algorithms.

The only thing we expect its result is a causal graph, optimal according to this criterion.

The model structure includes:

- 1) the time is continuous  $\frac{dx}{dt} = f(x, t)$  (basic: discrete with  $\Delta t = 1$ ),
- 2) the time series  $X, Y, Z$  has a spacial structure  $\frac{dx}{dx} = g(x, s)$  (basic: scalar-valued),
- 3) there is no limitations of the time as in the ESG,
- 4) the model is linear in the latent space (basic: linear in the observable space),
- 5) the latent space represent a spectrum of the observable domain.

The graph structure:

- 1) the graph is a bipartite graph between the observable and the latent,
- 2) the direct graph is isomorphic to the reverse graph,
- 3) the causal graph is in the hypergraph that groups the variables or the time.

The time structure for the convolutional model  $\int_{\tau=0}^T f(\tau)h(t-\tau)d\tau$

The causal graph  $G = G(t)$  is a time series. We suppose either:

- 1) it is stationary fot all  $t$ ,
- 2) it is stationary for  $t \geq T$  for with the period  $T$ ,
- 3) it is non-stationary but with rejection to discover its structure.

## 4 Appendix. Natural Gradient Descent

The natural gradient descent is a method to optimize the model parameters  $\theta$  by estimating the distribution

$$p(\theta) \propto \exp(-\mathcal{L}(\theta))$$

where  $\mathcal{L}(\theta)$  is the loss function of the model. The natural gradient descent is defined as

$$\theta_{t+1} = \theta_t - \eta F^{-1}$$

where  $F$  is the Fisher information matrix of the model parameters  $\theta$ .

## Fisher Metric and Natural Gradient

When working with probabilistic models, we can use a metric that is more natural for probability distributions: the Kullback–Leibler (KL) divergence. Suppose we have a likelihood model

$$p(x \mid \theta),$$

where  $x$  denotes fixed data and  $\theta$  is a parameter vector.

Let  $\theta_t$  denote the current parameter values, and suppose we wish to find updated parameters  $\theta_{t+1}$ . Under the natural gradient framework, this is done by solving the following optimization problem:

$$\theta_{t+1} = \arg \min_{\theta} [f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + D_{\text{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t))].$$

In general, this minimization problem is intractable. However, the KL divergence can be approximated locally using a second-order Taylor expansion. This expansion yields the Fisher information matrix  $F$ , which captures the local curvature of the statistical manifold. In particular, around  $\theta_t$ , we have

$$D_{\text{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \approx \frac{1}{2}(\theta - \theta_t)^\top F(\theta - \theta_t),$$

where the Fisher information matrix is defined as

$$F = \mathbb{E}_{x \sim p(x \mid \theta)} \left[ (\nabla_{\theta} \log p(x \mid \theta)) (\nabla_{\theta} \log p(x \mid \theta))^\top \right].$$

The Fisher information matrix therefore encodes the curvature of the likelihood-based loss function in parameter space. Using this approximation, the resulting natural gradient update rule becomes

$$\theta_{t+1} = \theta_t - \gamma F^{-1} \nabla f(\theta_t),$$

where  $\gamma > 0$  is the learning rate.

## Natural Gradient from an Information-Geometric Perspective

Consider a parametric family of probability distributions

$$\mathcal{M} = \{p(x \mid \theta) \mid \theta \in \Theta \subset \mathbb{R}^d\},$$

which forms a smooth statistical manifold  $\mathcal{M}$ . Each parameter vector  $\theta$  corresponds to a point on this manifold, and the coordinate system  $\theta$  is generally non-Euclidean.

The natural notion of distance between nearby probability distributions on  $\mathcal{M}$  is given by the Kullback–Leibler divergence. For two infinitesimally close distributions  $p(x \mid \theta)$  and  $p(x \mid \theta + d\theta)$ , the KL divergence admits the second-order expansion

$$D_{\text{KL}}(p(x \mid \theta + d\theta) \parallel p(x \mid \theta)) = \frac{1}{2} d\theta^\top G(\theta) d\theta + o(\|d\theta\|^2),$$

where

$$G(\theta) = \mathbb{E}_{x \sim p(x|\theta)} \left[ (\nabla_{\theta} \log p(x | \theta)) (\nabla_{\theta} \log p(x | \theta))^{\top} \right]$$

is the \*\*Fisher information metric tensor\*\* on  $\mathcal{M}$ .

Let  $f(\theta)$  be a smooth objective function defined on the manifold (e.g., the negative log-likelihood or empirical risk). Given the current point  $\theta_t \in \mathcal{M}$ , we seek a new point  $\theta_{t+1}$  by minimizing a first-order approximation of  $f$  subject to a constraint on the geodesic distance induced by the Fisher metric. This leads to the optimization problem

$$\theta_{t+1} = \arg \min_{\theta} \left\{ f(\theta_t) + \nabla f(\theta_t)^{\top} (\theta - \theta_t) \mid D_{\text{KL}}(p(x | \theta) \parallel p(x | \theta_t)) \leq \epsilon \right\}.$$

Using the local quadratic approximation of the KL divergence, this constrained problem reduces to minimizing the linearized objective under a Riemannian norm constraint. The solution is given by a step in the direction of the \*\*natural gradient\*\*, defined as

$$\tilde{\nabla} f(\theta) = G(\theta)^{-1} \nabla f(\theta).$$

Accordingly, the natural gradient descent update takes the form

$$\theta_{t+1} = \theta_t - \gamma \tilde{\nabla} f(\theta_t) = \theta_t - \gamma G(\theta_t)^{-1} \nabla f(\theta_t),$$

where  $\gamma > 0$  is the step size.

From an information-geometric standpoint, this update corresponds to moving along the direction of steepest descent with respect to the Fisher–Rao Riemannian metric, ensuring invariance under smooth reparameterizations of the statistical manifold.